

AREA

How to read the figures. — given --- auxiliary (added in the proof) — what is being proved or asked

10-1 INTRODUCTION

You already have an intuitive understanding of the concept of *area*. In this chapter we will describe *area* in much the same way as we described length in Chapter 5. There we determined a number that we called the length of a line segment by choosing an arbitrary *unit segment* and *counting*. Here, for a given geometric figure, we determine a number that we call the *area* of the figure by first choosing an arbitrary *unit square* and then *counting*.

It is much more difficult to describe the *counting process* for area than it is for length. We content ourselves with a crude description of the counting process as it relates to finding areas of rectangles. Once the area formula for rectangles is determined, we employ it to derive area formulas for other simple polygons.

10-2 AREA BY COUNTING

We understand by *polygon*, of course, the set of points on the sides. When we refer to the *area* of a *polygon*, we really are con-

sidering the set of interior points. It would be more descriptive to speak of the *area* of this *interior set*. But for brevity, we simply speak of the area of a polygon. First, we study rectangular polygons.

We need to explain what we mean by the area of a rectangle. It will help to consider a few examples.

Having chosen a unit length, a square 1 unit of length on a side is said to have 1 unit of area. Clearly, if we had two squares side by side, as in Fig. 10-1, we would want to count the area as 2. Suppose we had a rectangle with sides of lengths 3 and 2. Figure 10-1 shows that we would want the area to be 6 units.

If we had a square of side $\frac{1}{2}$ unit, then placing it in a unit square as in Fig. 9-2,¹ we see that the area should be only $\frac{1}{4}$ unit.

If we had a rectangle with sides $2\frac{1}{2}$ and 3 units, what would the area be?

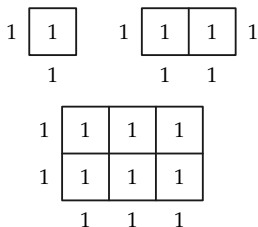


FIGURE 10-1

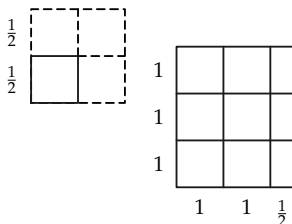


FIGURE 10-2

Exercise Group 10-1

¹*Transcriber's note.* The 1961 printing reads "Fig. 9-2" here; the figure meant is plainly Fig. 10-2. Reproduced as printed.

1. Draw a rectangle whose sides are 3 and 5 units. Determine its area by decomposing it into unit squares and counting.
2. Sketch a figure to illustrate why the area of an m by n rectangle is mn units. Here, m and n are positive integers.
3. A rectangle is $3\frac{1}{3}$ units long and 3 units wide. Find its area. What is the largest size square, such that the rectangle can be covered by an integral number of squares? Suppose that the rectangle were $3\frac{1}{3}$ by 2. What is the answer then?
- *4. Suppose that the length of a rectangle is a , and its width is b/c , where a , b , and c are positive integers. Obtain the area by dividing the rectangle into squares $1/c$ units on a side and then counting.
5. A rectangle has sides 2.2 and 3.3 in. Obtain its area by dividing it into small squares and counting.
6. A rectangle has sides 3.24 and 4.83 ft. Describe how to obtain the area by counting squares.
- *7. The sides of a rectangle are a/b and c/b units, where a , b , and c are positive integers. Show how its area is obtained by counting squares.

10-3 DEFINITION OF RECTANGLE AREA

In the previous section we saw how to obtain the area of a rectangle by counting squares if the rectangle has sides whose lengths are integers. Remember that, so far, we have not explicitly defined what area is, but have used only our previous experience.

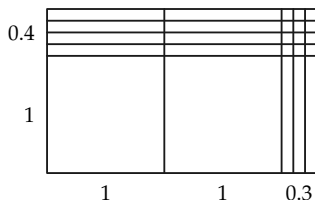


FIGURE 10-3

Suppose the lengths of the sides of a rectangle are l and w , where l and w are terminating decimals. If l and w end in 10ths, then we chop up the rectangle into squares 0.1 on a side. If l and w end in 100ths, we chop the rectangle into squares 0.01 on a side, and so on. In Fig. 10-3, we have started to chop up the rectangle, which is 1.4 by 2.3. Is it necessary to draw more of the small squares in order to count them?

It is rather obvious that the method of counting could be applied to a rectangle whose sides are any rational numbers.

Exercise 10-2

Find, by counting, the area of the rectangle whose sides are (a) 2.5 and 3.3, (b) 2.23 and 1.77, (c) $5/3$ and $7/3$, (d) $5/3$ and $1/2$, (e) $5/3$ and $3/4$.

In all our problems so far, we have found, by counting squares, that the area of the rectangle is lw .

Now suppose that at least one of the sides of the rectangle is not a rational number, that is, this side and the unit are not commensurable. To be definite, let us consider a rectangle whose length is $\sqrt{2}$ and whose width is 1. (Remember that $\sqrt{2}$ is not rational, and that *approximately* $\sqrt{2} = 1.4142$.) Examination of Fig. 10-4 shows that the area of the rectangle is greater than 1.4 and less than 1.5. We may see this by counting squares. If we were to subdivide the length into 100ths, we would also see that

the area is greater than 1.41 and less than 1.42. As we approximate to $\sqrt{2}$ more and more closely by rational numbers, for example, by 1.4, 1.41, 1.414, 1.4142, ..., we will get rectangles whose areas are ever closer to $\sqrt{2}$.

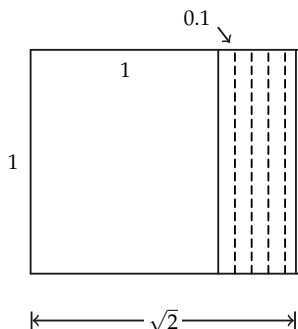


FIGURE 10-4

Therefore, the situation in which we find ourselves is as follows. If the sides of the rectangle are rational numbers, we can get the area by counting. If the sides are not rational, then simple counting does not give us the area. In this latter case, we must approximate to the sides by rational numbers and consider the *limit** of the areas of the rectangles whose sides have rational numbers for lengths. It is possible to prove that this limit exists and is equal to the product lw .

You see that to define the area of a rectangle by starting with counting may be quite troublesome. Therefore, in this book we do not give a detailed definition of area of a rectangle, but only state the following theorem, which tells us how to calculate the area.

*The word "limit" has a precise mathematical sense, which need not concern us now. Roughly, the idea is that the areas of the approximating rectangles get close to a number that we call the *area* of the given rectangle.

Theorem 10–1. *The area of a rectangle whose sides have lengths l and w is lw .*

Exercise Group 10–3

1. A square has side x yards. What is its area in square feet?
2. What is the area of a rectangle $3\sqrt{2}$ units long and $2\sqrt{2}$ units wide?

10–4 AREA OF POLYGONS

So far, the only polygons whose areas we have studied are rectangles. In discussing areas of other polygons, and in particular triangles, perhaps we should define area in terms of some kind of counting process. If we did, we would be forced to concern ourselves with limits, so we will determine areas of polygons by a method used by the Greek geometers. Our basic formula for the area of a rectangle will enable us to find areas of triangles, and other figures, without any counting of squares.

Remember that although the Greeks used numbers in commerce, they had no numbers other than integers in their geometry. They did not compute numerical measures of angles, lines, or areas. Just as in a primitive society a hunter might compare his son's height with that of the son of his friend by marking his spear, so the Greeks devised a method for comparing the sizes of polygons. Of course, the Greeks had no word in their geometry corresponding to our term "area," since area is a *number*. Instead of saying that two geometric figures had *equal areas*, they merely remarked that the figures were *equal*. The Greeks showed that two polygons were *equal* by cutting them up into congruent triangles. Note that the word *equal* meant *the same size* to the Greeks. This is different from the word's meaning in modern

mathematics, where *equal* means *identical*. Figure 10-5(a) illustrates what is meant. The Greek geometers would call rectangle $ADBE$ and isosceles triangle ABC *equal* because $\triangle ABE \cong \triangle ACE$ and $\triangle ABD \cong \triangle ACE$.

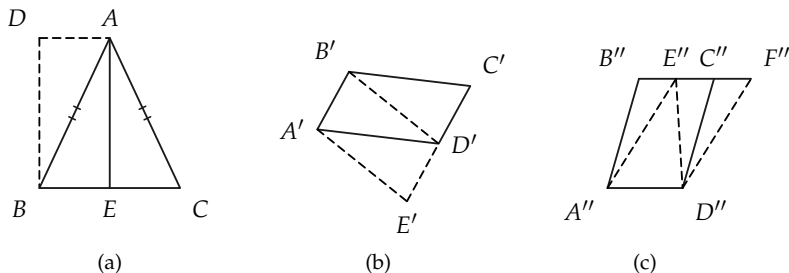


FIGURE 10-5

Exercise 10-4

In Fig. 10-5(b), $A'B'C'D'$ is a parallelogram, and $\overline{A'B'} \cong \overline{D'E'}$. Show that $A'B'D'E'$ and $A'B'C'D'$ are *equal* in the sense of Greek geometry. Do the same for parallelograms $A''B''C''D''$ and $A''E''F''D''$ in Fig. 10-5(c).

We use the methods of the Greek geometers to develop area formulas. But before doing so, we need a few theorems about areas of geometric figures. These theorems can be proved, using properties of numbers and our geometric postulates, but their proofs are long and difficult. Accordingly, we treat them as postulates even though proofs can be given.

Theorem 10-2. *Congruent triangles have equal areas.*

Theorem 10-3. *If two simple polygons have one side and only the points of this side in common, and if, moreover, they have no interior points in common, then the simple polygon formed by the sides that*

remain after the common side is deleted has an area equal to the sum of the areas of the original polygons.

Theorem 10-2 is intuitively obvious, since any method of counting squares to get the area of one triangle can be duplicated for the other triangle. Draw a figure to illustrate this. Figure 10-6 will make the meaning of Theorem 10-3 clear. The theorem tells us that the area of polygon $ABCDGFE$ is the sum of the areas of polygons $ABCDE$ and $DEFG$.

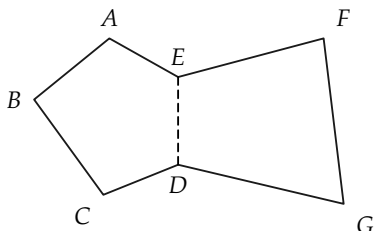


FIGURE 10-6

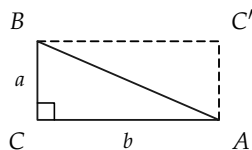


FIGURE 10-7

With these two theorems and our rectangle formula, we can establish theorems on the areas of triangles and parallelograms.

Theorem 10-4. *If a and b are the lengths of the legs of a right triangle, then the area of the triangle is $\frac{1}{2}ab$.*

Proof. Let the right triangle be $\triangle ABC$, as in Fig. 10-7, and let C' be chosen as indicated in the figure. (How is C' chosen?)

STATEMENTS	REASONS
1. $\triangle ABC \cong \triangle BAC'$.	Why?
2. Area $(\triangle ABC) = \text{area } (\triangle ABC')$.	Theorem 10-2.
3. Area $(\triangle ABC) + \text{area } (\triangle ABC') = \text{area } (\square AC'BC) = ab$.	Theorem 10-3 and formula for area of rectangle.
4. $2 \text{ area } (\triangle ABC) = ab$.	Statements 2 and 3.
5. Area $(\triangle ABC) = \frac{1}{2}ab$.	Statement 4.

In a parallelogram, if we call the length of one side the *base*, then the corresponding *altitude* (or *height*) is the perpendicular distance from this side to the line containing the opposite side. From Theorem 10-4, we can deduce the area of a parallelogram.

Theorem 10-5. *The area of a parallelogram is equal to the product of its base and its height.*

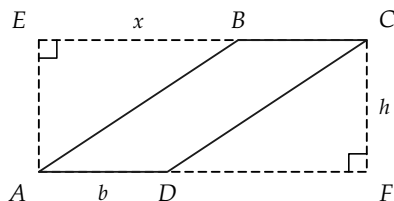


FIGURE 10-8

Proof. Let the parallelogram be $ABCD$, as in Fig. 10-8, its base $b = \overline{AD}$, and its height h . Consider, as in the figure, the rectangle $AECF$. (How can the vertices E and F be found?) Let $\overline{EB} = x$.

STATEMENTS	REASONS
1. $\triangle AEB \cong \triangle CFD$.	Why?
2. Area $(\triangle AEB) = \text{area}(\triangle CFD) = \frac{1}{2}hx$.	Why?
3. Area $(\square AE CF) = \text{area}(\triangle AEB) + \text{area}(\square ABCD) + \text{area}(\triangle CFD)$.	Why?
4. $\overline{AF} = \overline{EC} = b + x$.	Why?
5. $h(b + x) = \frac{1}{2}hx + \text{area}(\square ABCD) + \frac{1}{2}hx$.	Statements 3 and 4.
6. $hb + hx = \text{area}(\square ABCD) + hx$.	Statement 5 and algebra.
7. Area $(\square ABCD) = bh$.	Statement 6 and algebra.

We can now quite easily derive the familiar formula, $A = \frac{1}{2}b \cdot h$, for computing the area of a triangle. Here, b (base) is the length of a side of a triangle, and h (height) is the perpendicular distance from the opposite vertex to the line containing the base.

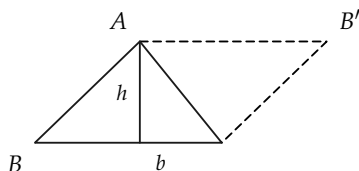


FIGURE 10-9

Theorem 10-6. *The area of a triangle is given by the formula $A = \frac{1}{2}b \cdot h$ (Fig. 10-9).*

The proof is left to the student.

From these last two theorems, we may conclude that parallelograms (or triangles) with equal bases and heights have equal

areas. The area formula for triangles provides us with a tool for determining areas of polygons in general, for the area of any polygon can be computed as the sum of the areas of certain triangles.

Exercise Group 10–5

1. A triangle has an area of 30 and a base of 6. Determine its height.
2. Two adjacent sides of a parallelogram have lengths of 12 and 8 in. These sides determine a 30° angle. Compute the area of the parallelogram.
3. A triangle has area 50 and altitude 8. What is the length of its base?
4. A rhombus has one angle of 45° and is 2 ft on a side. Find its area.
5. In Fig. 10–10, l and m are parallel and 3 ft apart. $\overline{AB} = \overline{CD} = 5$ ft. If the distance from B to C is 1 mi, what is the area of the quadrilateral $ABDC$?
6. What is the area of $\square ABCD$ if the sides are as shown in Fig. 10–11 and $BD \perp AD$?
7. The area of a triangle is A , and its base is x . Find the altitude in terms of A and x .
8. The base of a parallelogram can be expressed in terms of its area and height. What is this formula?
9. Two triangles have the same side for base. The height of one triangle exceeds that of the other by 4 in. The areas differ by 40 in². What is the length of the common base?
10. The midpoints of the sides of a parallelogram determine a polygon. Compare the area of this polygon to the area of the original parallelogram.
- *11. The midpoints A_1 , B_1 , and

C_1 of the sides of $\triangle ABC$ are joined to form a second triangle. Consider also the midpoints A_2 , B_2 , and C_2 of the sides of triangle $A_1B_1C_1$. This process is continued. Compare the area of $\triangle A_5B_5C_5$ with that of $\triangle ABC$.

- *12. Triangles ABC and $A'B'C'$ are similar. The lengths of two corresponding sides are s and s' , where $s = 5s'$. If A and A' represent the respective areas of the triangles, prove that $A = 25A'$. Generalize this problem, and establish the theorem that *if two triangles are similar, the ratio of their areas is equal to the ratio of the squares of the lengths of two corresponding sides*.
13. $\triangle ABC$ is similar to $\triangle A'B'C'$, and $AB = 2A'B'$. How do their areas compare? Draw a figure.
- *14. Plans for a house are drawn, so that $\frac{1}{4}$ in. on the plan corresponds to 1 ft in the house. If the plans cover an area of 105 in², how many square feet in the house?
15. Two equilateral triangles have sides of 8 and 10 in., respectively. How do their areas compare?
16. A rhombus has sides 10 in. long. If one angle is 30°, what is its area? if 60°? if 120°?
17. Find the area of an isosceles right triangle if one leg is 20 ft long.
18. In Fig. 10–12, $AD \parallel BC$. Compute the area of the figure.
- *19. Generalize Exercise 18, designating the lengths of the parallel sides, or bases, by b and b' and the height by h . Of course, you are developing a formula for the area of a trapezoid.
20. Besides the proof suggested by the dotted line in Fig. 10–12, Fig. 10–13 also suggests a proof. Here, it is assumed that $b > b'$. Make a proof.

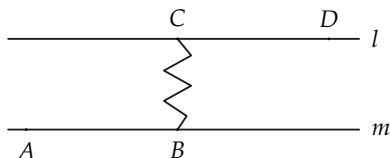


FIGURE 10-10

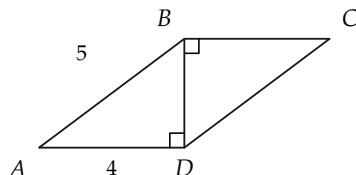


FIGURE 10-11

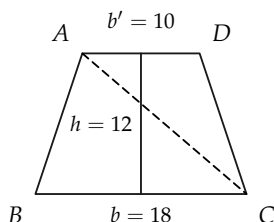


FIGURE 10-12

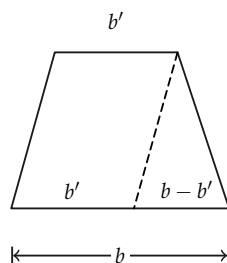


FIGURE 10-13

21. The bases of a trapezoid are 3 and 12, and the altitude is 6. What is its area?
22. The area of a trapezoid of height 6 in. is 39 in^2 . If one base is 6 in. what is the other?
23. One base of a trapezoid is twice as long as the other. If the area is 30 ft^2 and the altitude is 5 ft, what are the two bases?
24. An equilateral triangle has sides 18 in. long. Find its area in square feet.
25. An equilateral triangle has an area of $(9\sqrt{3})/4$. How long is its side?
26. In Fig. 10-14, $ABCD$ is a parallelogram. Prove that the four small triangles, $\triangle AOB$, $\triangle BOC$, etc., all have equal areas.
27. The areas of two similar triangles are 360 and 250 square units. If a side of the first triangle is 8 units, how long is the corresponding

side of the second triangle?

and the sides 10 in.

- *28. Prove that the ratio of the areas of similar polygons is equal to the square of the ratio of corresponding sides.

29. Find the area of a regular hexagon 6 in. on a side. What will be the area of a regular hexagon whose side is 8 in.?

- *30. If a , b , and c are the lengths of the sides of any triangle (Fig. 10-15), and $s = \frac{1}{2}(a + b + c)$, then the area is given by

$$A = \sqrt{s(s-a)(s-b)(s-c)}.$$

This is "Heron's formula," named after an Alexandrian Greek by that name of about 100 B.C., possibly later. Show that this formula is correct.

31. The sides of a triangle are 8, 11, and 13 in. Find its area.
32. The sides of a triangle are 17, 25, and 26 ft. Find its area. Find the height to the 25-ft side.
33. Find the area of an isosceles triangle if the base is 12 in.

34. Find the area of an equilateral triangle if one side is 6 in.

35. If, in Fig. 10-16, $ABCD$ is a parallelogram, with $\overline{BE} = 2$ and $\overline{ED} = 6$, what is the number

$$\frac{\text{Area}(BEC)}{\text{Area}(ABCD)}?$$

36. A square has a diagonal of length 12 in. Compute the area of the square. Determine a formula for the area of a square in terms of the length d of its diagonal.

37. The diagonals of a rhombus are 10 in. and 6 in., respectively. What is its area? Generalize this problem by formulating and proving a theorem.

38. In Fig. 10-17, $\angle C$ is right, $\overline{CD} \perp \overline{AB}$, $\overline{AC} = 5$, and $\overline{CB} = 12$. Compute the areas of triangles ABC , ACD , and BCD . Check your work by verifying that the sum of the area of the latter two triangles is equal to the area of $\triangle ABC$.

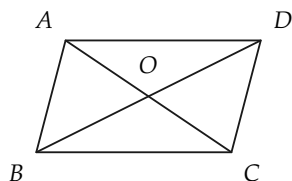


FIGURE 10-14

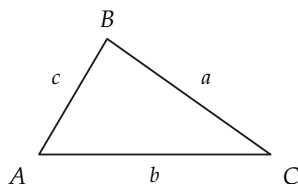


FIGURE 10-15

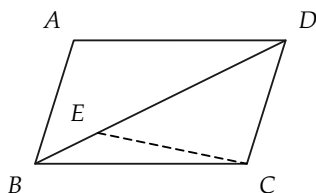


FIGURE 10-16

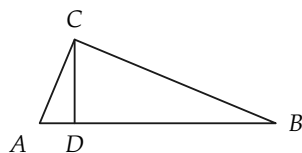


FIGURE 10-17

10-5 AREA PROOF OF THE PYTHAGOREAN THEOREM

The Greek geometers thought of the Pythagorean theorem, not as a numerical relationship between lengths of the sides of a right triangle, but rather as a relationship between areas. Viewed in this light, the theorem may be stated as:

Theorem 10-7. *For every right triangle, it is true that the square on the hypotenuse has an area equal to the sum of the areas of the squares on the other two sides (see Fig. 10-18).*

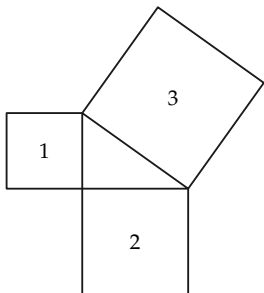


FIGURE 10-18

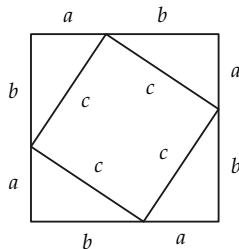


FIGURE 10-19

$$\text{Area } 1 + \text{area } 2 = \text{area } 3.$$

Probably more different proofs have been given for the Pythagorean theorem than for any other theorem in mathematics. Our earlier proof did not refer to area at all. It was essentially an “algebraic style” proof. We give a proof below that is a kind of algebraic and area proof. If the sides of the right triangle are a , b , and c , consider Fig. 10-19.

Proof. Evidently, the area of the large square, of side $a + b$, is equal to the area of the square of side c plus the areas of the four triangles. That is,

$$(a + b)^2 = c^2 + 4\left(\frac{1}{2}ab\right).$$

But $(a + b)^2 = a^2 + 2ab + b^2$, so that we get

$$a^2 + 2ab + b^2 = c^2 + 2ab.$$

From algebra then,

$$a^2 + b^2 = c^2.$$

There are many more proofs of this theorem. Your teacher can show you some if you are interested.

Exercise Group 10-6

1. Triangles ABC and $A'B'C'$ are similar. $\overline{AB} = 3$, $\overline{A'B'} = 5$, and $\text{area } (ABC) = 18$. Compute $\text{area } (A'B'C')$.
2. Determine the area of an equilateral triangle if the length of one side is 10.
3. Show that the area of an equilateral triangle is given by the formula $A = (s^2/4) \cdot \sqrt{3}$, where s is the length of a side.
4. Develop a formula for the area of a regular hexagon if each side has length c .
5. The lengths of two adjacent sides of a parallelogram are 12 in. and 8 in. The included angle is 45° . Determine the area of the parallelogram.
6. The altitude of a rectangle is 12 ft. A diagonal has length 20 ft. Compute the area of the rectangle.
- *7. A square and a regular hexagon have equal perimeters. Find the ratio of their areas.
- *8. It is said that the Hindu mathematician Bhaskara, born about 1100 A.D., discovered a proof of the Pythagorean theorem and presented the figure like that in Fig. 10-20 with the single word, "Behold." Give a proof. In Fig. 10-20 $DF \parallel AC$, and $EG \parallel BC$. [Hint: $\overline{FC} = a - b$.]

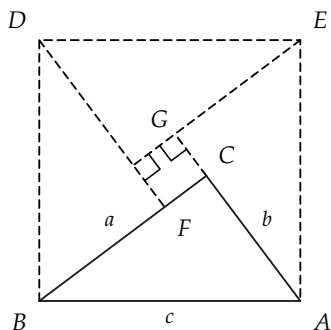


FIGURE 10-20

*10-6 CONSTRUCTION PROBLEMS

There are many RC constructions related to area concepts. It is interesting to note that it is possible to construct a triangle whose area is equal to the area of any given polygon. We indicate in Fig. 10-21 a procedure for constructing a triangle equal in area to a quadrilateral. DE has been drawn parallel to diagonal AC . Show that $\text{area}(ABE) = \text{area}(ABCD)$. [Hint: Show that $\text{area}(ADC) = \text{area}(AEC)$.]

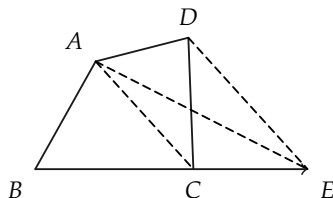


FIGURE 10-21

Exercise Group 10-7

1. Construct a triangle whose area is equal to that of pentagon $ABCDE$ (Fig. 10–22). Use the method employed above to construct a triangle equal in area to a given quadrilateral.
2. Show how to construct a square whose area is equal to the sum of the areas of two given squares.
3. Solve Exercise 2 with “sum” replaced by “difference.”
4. Construct a square whose area is $\frac{3}{4}$ of the area of a given square.
5. Construct a square whose area is equal to the area of a given rectangle.
6. Construct a rectangle whose area is equal to the area of a given triangle.
7. Construct a square whose area is the same as the area of a given triangle.
8. Construct a triangle similar to a given triangle and having exactly twice the area of the given triangle.
9. Show how to construct a polygon of $n - 1$ sides whose area is the same as that of an n -gon. Assume $n > 3$.
- *10. Prove that if P is any point in the interior of an equilateral triangle (Fig. 10–23), the sum of the lengths of the perpendiculars from P to the sides of the triangle is equal to the length of an altitude of the triangle. [*Hint*: Consider the areas of triangles APB , APC , and BPC .]

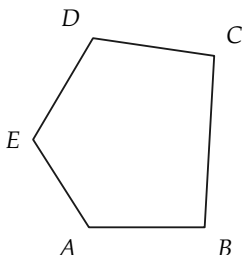


FIGURE 10-22

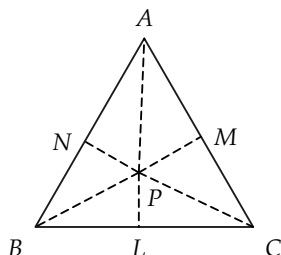


FIGURE 10-23

REVIEW OF CHAPTER 10

1. In this chapter we have studied methods for finding the areas of rectangles, triangles, and polygons. All area formulas were derived from the formula for the area of a rectangle.

Area $\square$ = length $\cdot$ width = lw .

Area $\square$ = base $\cdot$ altitude = bh .

Area $\triangle$ = $\frac{1}{2}$ base $\cdot$ altitude = $\frac{1}{2}bh$.

Area $\triangle$ = $\frac{1}{2}$ altitude $\cdot$ sum of bases = $h(b + b')/2$.

2. Review how the above area formulas were obtained.
3. Areas of similar figures are proportional to the squares of corresponding sides.

Review Exercises

1. $\triangle A_1B_1C_1 \sim \triangle A_2B_2C_2$. If area $(A_2B_2C_2) = 72$, and $\overline{A_1B_1}/\overline{A_2B_2} = \frac{3}{4}$, find area $(A_1B_1C_1)$.
2. Prove that a median of a triangle divides the triangle into two equal areas.

- *3. Prove that the medians divide a triangle into six triangles of equal areas (Fig. 10-24).
4. Prove that the area of a square is half the area of a square whose side is the diagonal of the given square.
5. Two similar triangles have areas of 108 and 48 ft². The first has altitude $7\frac{1}{2}$ ft; find the altitude of the second.
6. Prove that the area of a trapezoid is equal to the altitude times the length of the segment joining the midpoints of the non-parallel sides.
7. A triangle has one angle of 30° . Prove that the area is $\frac{1}{4}$ of the product of the lengths of the sides adjacent to the 30° angle.
8. In $\triangle ABC$, if D is the foot of the perpendicular from C to AB , prove that $\overline{AC}^2 - \overline{BC}^2 = \overline{AD}^2 - \overline{BD}^2$.
9. The hypotenuse of a right triangle is 41 ft. One leg is 40 ft. Find the length of the other leg.
10. The median of a trapezoid is 8 in., and one of the bases is 12 in. How long is the other base? If the area is $\frac{1}{2}\text{ft}^2$, what is the altitude?
11. In a right triangle, the altitude from the right angle is 14 in. From the foot of the altitude to a vertex is 8 in. How long is the hypotenuse?

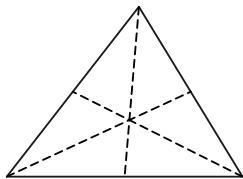


FIGURE 10-24

Review Test

1. What is the area of a parallelogram with sides 10 and 8 and an included angle of 30° ?
2. What is the area of a rhombus with side 10 if an angle is 45° ?
3. Find the area of an isosceles right triangle whose legs are 20.
4. One of the bases of a trapezoid is 15, and its height is 6. Find the other base if the area is 75.
5. Two similar triangles, T and T' , have two corresponding sides 6 and 15, respectively. What is the ratio of the area of T to the area of T' ?
6. The length of a diagonal of a square is 15. Find the area.
7. Two similar polygons have areas of 250 and 160. If the length of a side of the first is 6, what is the length of the corresponding side of the second?
8. Two triangles are similar, and the lengths of the sides of the first triangle are 3, 5, and 6. If the length of the greatest side of the other triangle is 15, compute the lengths of the other sides.
9. If the legs of a right triangle are in the ratio of 5 to 4, find the ratio of the segments formed on the hypotenuse by the altitude to the hypotenuse.
10. If one leg of a right triangle has length 25, and the altitude to the hypotenuse is 7, what is the area of the triangle?